

Wentao Zhou

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Multimodal learning · Reinforcement learning · Efficient 3D perception

Education

Georgia Institute of Technology Ph.D. student, Computer Science	Aug. 2026–Present
University of Virginia M.S., Computer Science	Aug. 2024–Apr. 2026
University of Michigan B.S., Computer Science (dual degree)	Aug. 2022–Apr. 2024
Shanghai Jiao Tong University B.S., Electrical and Computer Engineering (dual degree)	Sep. 2020–Aug. 2024

Selected Research & Publications

*Equal contribution.

Mutually Adversarial Self-Training with Evolving Data for Unified Multimodal Models

Wentao Zhou*, Weijie Gan*, Jiayun Wang

arXiv preprint, 2026 | MATE

- GRPO-based post-training alternates generation and understanding as challenger and solver; hard, source-consistent outputs form evolving training pools without a separate adversary.
- On Janus-Pro-1B, improved GenEval by **2.4 points**, DPG-Bench by **1.7 points**, and average performance across nine understanding benchmarks by **0.7 points** over the base model.

Recurrent Point Transformer via Test-Time Training

Wentao Zhou, Xuweiyi Chen, Charles Saunders, Aruni RoyChowdhury, Zezhou Cheng

NeurIPS 2026 | PointTTT

- Designed a recurrent Point Transformer V3 encoder with residual Large Chunk Test-Time Training layers in the encoder and decoder for 3D and 4D semantic segmentation.
- Processed large point clouds in fixed-size chunks with compact fast-weight states, supporting online full-sequence inference with nearly constant memory.

Stereo4DWalker: Learning 4D-aware Embodied Urban Navigation from Internet Stereo Videos

Wentao Zhou, Xuweiyi Chen, Vignesh Rajagopal, Jeffrey Chen, Rohan Chandra, Zezhou Cheng

IROS 2026 | Code

- Integrated stereo vision and explicit mid-level visual representations for navigation in dynamic urban environments, with automatically generated trajectory annotations.
- Used **30 hours of monocular training video**, 1.5% of CityWalker’s 2,000-hour corpus; introduced a **60-hour stereo walking dataset**.

WildRayZer: Self-supervised Large View Synthesis in Dynamic Environments

Xuweiyi Chen, Wentao Zhou, Zezhou Cheng

CVPR 2026 — Highlight | Code

- Self-supervised novel view synthesis separates static structure from transient motion, enabling transient-object removal and full-frame view synthesis in a single feed-forward pass.

Point-MoE: Large-Scale Multi-Dataset Training with Mixture-of-Experts for 3D Semantic Segmentation

Xuweiyi Chen, Wentao Zhou, Aruni RoyChowdhury, Zezhou Cheng

ICLR 2026 | Code

- Joint indoor/outdoor training achieved **70.8 mean mIoU** versus 67.2 for PTv3-L, using a specialized mixture-of-experts with **59M versus 97M active parameters** and no dataset labels.